

Aditya Shibu

GitHub: github.com/adityashibu
Portfolio Website: <https://adityashibu.vercel.app/>

Email: adityashibuonline@gmail.com
Mobile: +971 54 354 7825
LinkedIn: [linkedin.com/in/adityashibu](https://www.linkedin.com/in/adityashibu)
Dubai, UAE

EDUCATION

- Bachelor's of Science (Hons.) - Computer Science with specialization in AI** **GPA: 4.0/4.0**
Heriot Watt University, Dubai, UAE *Sept 2023 - Present*

RESEARCH AND PUBLICATIONS

- Simulating an Autonomous System in CARLA Using ROS2** **Research Paper** - Published
International Conference on Signal Processing and Information Security (ICSPIS) *November 2025*
- SkySim: Natural Language Control of Drone Swarms using LLMs** **Research Paper** - Published
International Conference on Drones and Unmanned Systems (DAUS) *February 2026*
- LiZIP: A Deep Learning Model for Compression of LiDAR point clouds** **Research Paper** - Preprint
School of Mathematical and Computer Sciences - Heriot Watt University *Target - RA-L*

WORK EXPERIENCE

- Lab for AI and Robotics Research (LAIRR) - Heriot Watt University** **Dubai, UAE**
Research Intern *Nov 2025 - Present*
 - Engineered **LiZIP**, a novel auto-regressive neural compression architecture for LiDAR point clouds, writing the full inference pipeline from scratch in C++, custom CUDA kernels, and TensorRT.
 - Optimized for bare-metal edge deployment on the NVIDIA Jetson AGX Orin, achieving real-time inference latency of **16.7ms (nuScenes) to 26.8ms (Argoverse)** per frame, while consuming **<20% CPU and <1% GPU utilization**.
 - Successfully outperformed industry standards **LASzip** and **Google Draco** across both datasets in compression efficiency, reducing payload sizes to **186 KB and 605 KB** respectively, while maintaining a near-lossless maximum error of **<0.017 mm**.
- AimbeyonD - Heriot Watt University** **Dubai, UAE**
Research Intern *Nov 2025 - Present*
 - Developing framework for control of QDrone2 in a motion capture environment using ROS2 and Gazebo, integrating with Ollama for on-device SLM inference to enable low-latency command processing
 - Developed **SkySim**, a ROS 2 Jazzy-based simulation environment on Gazebo for Natural Language Control of Drone Swarms using Large Language Models, enabling translation of high-level commands into low-level drone control instructions
 - Used **Gazebo** as the simulation environment for simulation translation and testing of features such as Artificial Potential Fields-based collision avoidance and formation flying
 - Implemented **ROS 2 Humble** nodes for drone swarm communication and coordination, integrating with **Ollama** for on-device SLM inference to ensure low-latency command processing
 - Deployed on **10x Bitcraze 2.1 Brushless Drones**, and using **Optitrack** as the motion capture system for streaming drone positions to SLM
- Atlas Racing FS** **Dubai, UAE**
Head of Driverless & Lead Autonomous Systems Developer *May 2025 - Present*
 - Developed end-to-end autonomous vehicle simulation pipeline in **CARLA**, implementing **deep neural networks for perception** and collecting **11.5K+ training images** from FOSCO dataset for real-time inference optimization
 - Engineered path planning algorithms (**Pure Pursuit, MPC**) integrated with **ROS 2 Humble**, achieving seamless simulation-to-reality transfer for Formula Student AI competition
 - Built comprehensive perception pipeline using **ZED2i stereo cameras and RoboSense Helios 16 LiDAR**, implementing **DBSCAN clustering** and sensor fusion for object detection and localization
 - Optimized **YOLO inference with TensorRT acceleration**, reducing latency by **88%** (140ms → 16ms) and achieving stable **40-50 FPS** for real-time autonomous vehicle operation
 - Led technical development resulting in **1st Place Overall Statics** and **1st Place Real World AI**, demonstrating production-ready autonomous systems performance
- Center for AI and Robotics (CAIR) - New York University** **Abu Dhabi, UAE**
Research Collaborator *Dec 2025 - Feb 2026*
 - Architected and trained a **Conditional Imitation Learning (CILRS)** neural network using PyTorch to navigate complex urban intersections based on high-level GPS commands.
 - Processed and synchronized multi-view RGB camera streams for end-to-end driving models
 - Configured and managed headless CARLA simulations on dual **NVIDIA L40s HPC** infrastructure
 - Developed Python data collection pipelines to generate synthetic urban driving datasets in CARLA (RGB + Telemetry).

PROJECTS

- **SkySim**: ROS 2 Jazzy based simulation environment on Gazebo for Natural Language Control of Drone Swarms using Large Language Models | **Tech: ROS2 Jazzy, Gazebo, Python, Ubuntu 24.04** (January 2025) | **To be released soon**
- **Hydrakon V1 and V2**: Worked with a team of 9 to build an autonomous racing car (Simulation and Hardware Integration) for Formula Student UK | **Tech: CARLA, YOLO, ROS2, Cartographer, Python, C** (April 2025) | <https://github.com/Atlas-Racing>
- **Local RAG**: Developed a fully local RAG with Ollama and LangChain | **Tech: Python, Ollama, and LangChain** (January 2025) | <https://github.com/jonathanjthomas/GDG-RAG-Demo>
- **Space Debris Classification**: Developed a space debris classification model using YOLO, enhancing training data by combining datasets from Roboflow for broader coverage | **Tech: Python, Jupyter Notebook, YOLO** (December 2024) | <https://github.com/adityashibu/spaceDebrisDetection>

AWARDS AND CERTIFICATIONS

- **SIMULINK Onramp** - Mathworks - March, 2026
- **MATLAB Onramp** - Mathworks - March, 2026
- **A Beginner's Guide to Autonomous Robots** - NVIDIA - March, 2026
- **Learning Linux Command Line** - LinkedIn - February, 2026
- **Babcock FS-AI Overall Statics Award Winner** - IMechE - July, 2025
- **FS-AI Real World Award Winner** - IMechE - July, 2025
- **Deep Learning Essentials Intermediate** - IBM - January, 2025
- **GitHub Foundations certification** - GitHub - December, 2024

LEADERSHIP AND VOLUNTEERING EXPERIENCE

- **Heriot Watt University** Dubai, UAE
MACS Department Officer *August 2025 - Present*
 - Lead and coordinate a team of **12 class representatives** across all Mathematical and Computer Sciences courses (Years 1-4), facilitating effective communication and problem resolution, serving as primary student advocate in faculty committees and departmental meetings, successfully addressing academic concerns and implementing policy changes that impact **600+ MACS students**
- **Google Developer Groups** Dubai, UAE
Lead of AI and ML *May 2025 - Present*
 - Promoted from Co-lead to Lead after organizing a successful workshop with **120+ attendees**, demonstrating exceptional event management and technical presentation skills
- **WattBuds - Peer Mentoring program** Dubai, UAE
Event Coordinator *September 2024 - March 2025*
 - Co-organized record-breaking campus event with **215+ attendees**, surpassing all previous university events and setting new benchmark for student participation and engagement
- **COP28** Dubai, UAE
IT Volunteer *December 2023*